Vamps, a Vegetation-AtMosphere-Plant-Soil water model

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Note on version 1.0

Since version 1.0 the S-Lang scripting engine has been replaced by embedded CPython. User scripts are now plain Python 3 files hooked in via `[vamps] xtrapy = myscript.py`. The S-Lang intrinsics described in the remainder of this manual are no longer available.

A Python package `vampspy` is provided for driving Vamps entirely from Python — no subprocess invocation, no temporary files. It supports single-column runs (via `Model.run()` or `Model.run_stepwise()`) and parallel multi-column / 2-D grid runs (`Model.run_grid()`). See `docs/vampspy_interface.md` in the source tree for the full API reference and `notebooks/` for worked examples.

Source and issue tracker: https://github.com/jaapschellekens/vampspy


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 * Parts of this program with permission derived from swap/swatr.
 *
 *    Author : Jan G. Wesseling
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Contents

1 Introduction **13**

I Model description **19**

2 Introduction **21**

3 The modules **23**
3.1 Introduction 23
3.2 The atmosphere 23
3.2.1 Potential evaporation 23
3.2.2 Actual evapotranspiration 25
3.3 Plants and trees 25
3.3.1 Interception 26
3.4 The soil 29
3.4.1 Introduction 29
3.4.2 Solving Richard's equation 29
3.4.3 Relation between Theta, ψ and K_{unsat} 30
3.4.4 Bottom boundary conditions 34

II User's guide **35**

4 Installation of Vamps **37**

5 Program organization **39**
5.1 Introduction 39
5.2 System requirements 39
5.3 Defaults file 40
5.4 Time series and time steps 40
5.5 The principle of layers in the soil section 41

6 The input file **43**
6.1 introduction 43
6.2 Input file format 43
6.3 Available sections and variables 44
6.3.1 [vamps] 44

6.3.2 [time] . . . 45
6.3.3 [run] . . . 45
6.3.4 [xout] . . . 45
6.3.5 [determine] . . . 45
6.3.6 [pevaporation] . . . 46
6.3.7 [evaporation] . . . 46
6.3.8 [interception] . . . 46
6.3.9 [canopy] . . . 46
6.3.10 [roots] . . . 47
6.3.11 [ts] . . . 48
6.3.12 [soil] . . . 49
6.3.13 [drainage] . . . 51
6.3.14 [*soilsectionname*] . . . 52
6.3.15 [layer_*n*] . . . 53
6.3.16 [fit] . . . 54
6.3.17 Valid datasets . . . 54
6.4 Interception . . . 55

7 Running Vamps 57
7.1 Command line options . . . 57
7.2 Manipulation of time series . . . 58
7.3 graphically displaying Vamps results . . . 58
7.4 Troubles.. . . . 58
7.4.1 Variable missing . . . 58
7.4.2 Mass-balance error warning . . . 58

8 Using S-Lang with Vamps 59
8.1 Introduction . . . 59
8.2 How Vamps interfaces with **S-Lang** . . . 59

9 Vamps specific functions and variables 61
9.1 Intrinsic functions (programmed in C) . . . 61
9.2 User defined functions (written in **S-Lang**) . . . 61

III Case studies 63

10 Introduction 65

11 A *Pinus caribaea* plantation forest 67
11.1 Introduction . . . 67
11.2 Results . . . 67

List of Figures

1.1 Simplified diagram of the flow of water through a forested ecosystem (after Bruijnzeel (1983)) 15
1.2 Simplified flow diagram of the Vamps model 16

5.1 Relation between the layers variable in the soil section and the amount of physical layers. 42

11.1 Measured (plus signs) and modeled (line) water content for layers at 6, 22 and 42 cm. below the surface 69
11.2 Measured (plus signs) and modeled (line) water content for layers at 80 and 104 cm. The bottom figure shows measured and modeled average water content. 70
11.3 71
11.4 72

List of Tables

1.1 Selected features of the Vamps model 14

3.1 Representative values for hydraulic parameters. After Clapp (1978). . 33

11.1 Soil physical parameters from to Tulasewa site used as input for Vamps 67

11.2 Summary of modeled water balance results for the period July 2 – September 7 68

List of Examples

3.4.1 The C code used by Vamps to estimate the maximum dt 31

Chapter 1

Introduction

Vamps is a one-dimensional water balance model for soils covered by forest, agricultural crop or bare land particularly suited for use in tropical environments. Vamps can be used as a complete forest hydrological model or as a tool to determine just one or more parameters (e.g. calculate interception using Gash' model). Vamps was designed to be extensible. At the same time the model should be simple to operate. The model is therefore build in a modular fashion and is equipped with a simple C-like scripting language. Features such as variable time steps and the ability to select modules allow the model to be used in both data-rich and data-poor environments.

The model has been developed as part of a collaborative project 'Hydrological modeling in a humid tropical island setting: with special reference to The Luquillo experimental Forest, Puerto Rico' between the International Institute of Tropical Forestry and the Vrije Universiteit Amsterdam (Bruijnzeel and Scatena 1994). The model is free and distributed with complete source code. It runs on a variety of operating systems[1].

The soil part of Vamps was developed after SWATR[2] (Feddes, Kowalik and Zarasny 1978). The large number of models that have been developed is an indication that models need to be tailored or adapted for each specific application. In order to make Vamps easily extensible a C-like interpreted language (Davis 1995b, Davis 1995a) allows the user to redefine builtin functions with functions tailored to the site in question. These features in combination with testing on several tropical sites make the model of particular importance to those interested in the tropics as a whole and tropical forests in particular.

The main development goals of Vamps are: *(i)* make a parameterized model which can adequately describe the flow of water in a forested environment (on a plot base) and *(ii)* be flexible enough to be applied to the variety of forested environments that exist. Due to its flexibility it is also possible to apply Vamps to plots covered with agricultural crops or bare soils. Furthermore the software is distributed with complete source code and documentation to allow the user to adapt the program to new situations and examine the program's internal operation.

[1] At the moment Vamps runs on several operating systems including UNIX (Sun-OS, Linux, Net-BSD and AIX) MS-DOS and OS/2

[2] Currently the swap/swatr program is maintained and documented in by: Dept. of Agrohydrology, Winand Staring Centre and Dept. of Water Resources, Wageningen Agricultural University

- The program is written in ANSI C (Kernighan and Ritchie 1988)
- Documentation for both model principles and model operation
- Both binaries and source code available via the World Wide Web
- Most parts of the model can be used separately
- Variable time-steps (e.g. 0.5 seconds to two weeks)
- No limit to the number of soil layers
- Run with Sun-OS, Linux, AIX, Net-BSD, MS-DOS and OS/2
- Includes pre- and postprocessors
- Simple ASCII input and output file format
- Easy extension by the users through builtin (high level) library functions
- Simple to adjust to specific needs at source code level

Table 1.1: Selected features of the Vamps model

Model principles

The flow of water through a forested ecosystem is shown in Figure 1.1. Three pathways are generally distinguished by which precipitation reaches the forest floor. A small fraction of the precipitation reaches the forest floor without touching leaves or stems. This is known as direct throughfall. Another small fraction flows down the tree trunks as stemflow. The remaining fraction hits the forest canopy and will leave the canopy as crown drip or evaporate from the wet canopy depending on canopy storage capacity and shape, the kinetic energy of the droplets and the atmosphere's evaporative demand. Water infiltrating the soil profile can runoff laterally as saturated or unsaturated flow, percolate to deep groundwater or is extracted by the roots of plants or trees. If the troughfall intensity exceeds the infiltration capacity of the topsoil, Hortonian overland flow can occur. The top layer of the soil profile can become saturated resulting in saturation overland flow. Vamps simulates most of these fluxes which is illustrated in Figure 1.2. The model is mainly driven by input from the atmosphere[3].

Most of the fluxes in the model can either be calculated or given by the user. The atmosphere, canopy and soil part of the model can be combined or used separately. However, the strength of the model is the ability to combine these parts into an integrated model for the forest hydrological cycle.

The following briefly summarizes the methods used to calculate the water fluxes:

Throughfall can be determined in several ways. *(1)* using a complete canopy water balance model based on the model developed by Rutter, Kershaw, Robins and

[3]As the bottom boundary conditions can also change in time one could probably regard this as a driving force as well

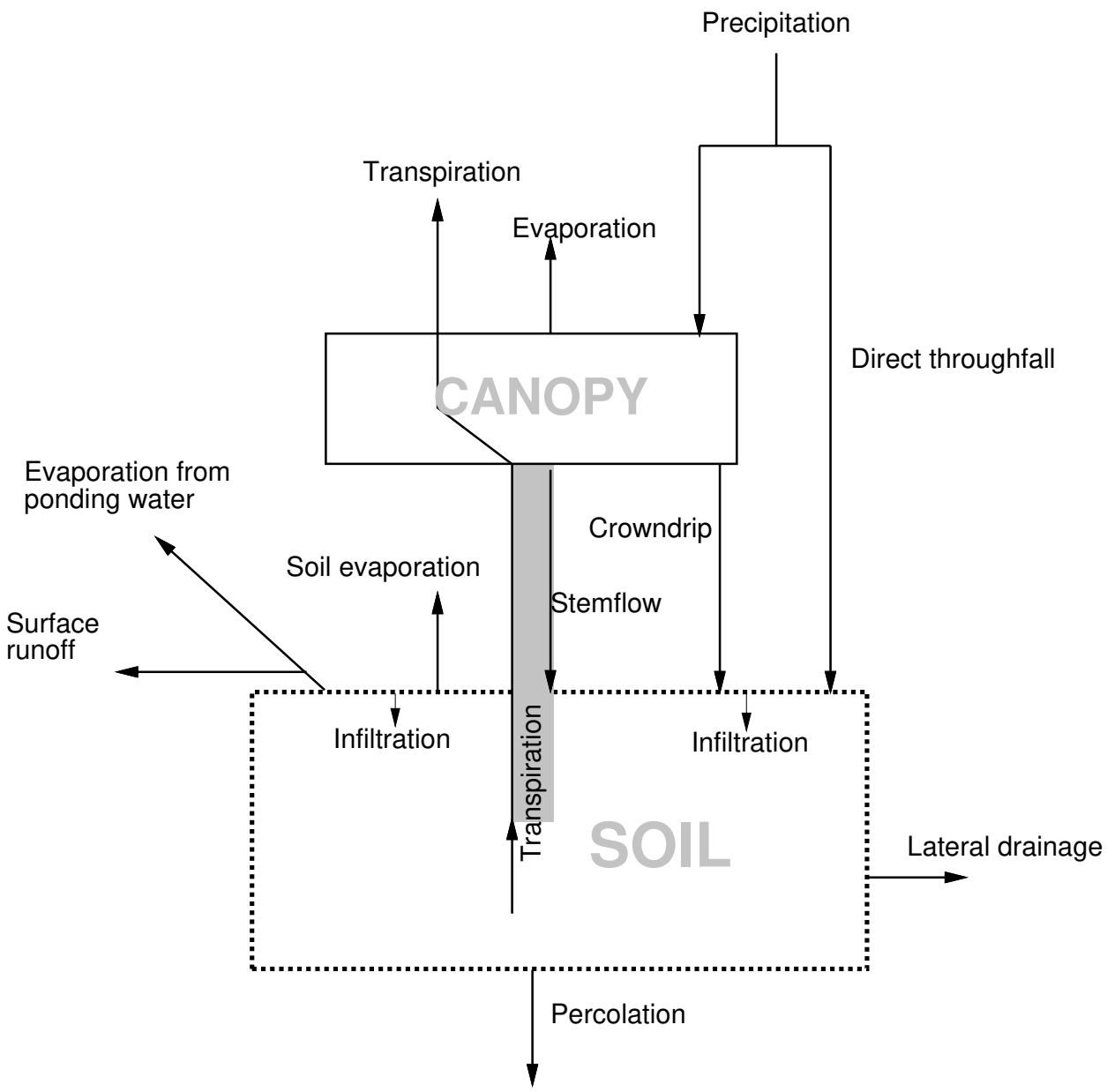


Figure 1.2: Simplified flow diagram of the Vamps model

Morton (1971). *(2)* using one of the simpler analytical approaches by Gash (1979) or Calder (1986). *(3)* the Leaf Area Index based solution also used by the TOPOG model (Vertessy, Hatton, O'Shaughnessy and Jayasuriya 1993).

Transpiration is commonly determined using the Penman-Monteith equation (Monteith 1965). Several alternative methods can be used as well (Penman 1956, Makkink 1957, Makkink 1961, Commissie Voor Hydrologisch Onderzoek TNO 1988).

Soil water fluxes in the unsaturated zone are determined using Richard's equation. An adapted form of the numerical solution described by Feddes et al. (1978) and Belmans, Wesseling and Feddes (1983) is used. The relation between water content and pressure head can be described in several ways (van Genuchten 1980, Mualem 1976, Clapp and Hornberger 1978).

These fluxes can also be calculated by a user supplied method.

Other information

The latest version and other information is available on the Word Wide Web (http://flow.geo.vu.nl).

Part I

Model description

Chapter 2

Introduction

This part of the manual will describe the principles on which Vamps was based. If things are not explained here a reference to existing literature is given. If you think the manual is incorrect it might be out of date as Vamps is still being developed. The best thing is to check the source code if you have any doubts.

The main goals when the development of Vamps started were: *(1)* make a parameterized model which can adequately describe the flow of water in a forested environment (on a plot base) and *(2)* be flexible enough to be able to be applied to the variety of forested environments that exist. However, because Vamps is very flexible it is also possible to apply Vamps to plots covered with agricultural crops or bare soils. Also *(3)* it is the authors opinion that (modelling) software should be distributed with complete source code and documentation to allow the user to adapt the program to any new situation and examine the program's internal workings.

The flexible nature of Vamps can be a bit confusing at first because of the large amount of choices to be made. However, this manual and the example files should be able to guide you to a successful application of the Vamps model.

The flow of water through a forested ecosystem is described in Figure 1.1. In general three pathways can be distinguished by which precipitation reaches the forest floor. A small fraction of the water reaches the forest floor without touching leaves or stems – known as direct throughfall. Another – also small – fraction of the water flows down the tree trunks as stemflow. The rest of the water hits the forest canopy and will leave this canopy as crown drip or evaporation from the wet canopy depending on canopy storage and shape, kinetic energy of the droplets and the athmosphere's evaporative demand. Water that infiltrates in the soil profile can leave this laterally as saturated or unsaturated flow, percolate to deep groundwater or it can be extracted by the roots of plants or trees. If the troughfall intensity exceeds the infiltration capacity of the topsoil Hortonian overland flow can occur. The top layer of the profile can become saturated resulting in Saturation overland flow. Vamps simulates most of these fluxes as can be seen in Figure 1.2 which shows a simplified flow diagram of the Vamps model. The model is mainly driven by input from the atmosphere[1]. In the following chapters the workings of Vamps and how the model simulates each of these fluxes will be explained.

[1] As the bottom boundary conditions can also change in time one could probably regard this as a driving force as well

It should be noted that most fluxes in the model can either be calculated or given by the user. The atmosphere, canopy and soil part of the model can be combined or used separately. However, the strength of models like Vamps is their ability to combine all these to an integrated model of the forest hydrological cycle. The canopy part in particular is closely interwoven with the atmosphere module and is hard to use seperate from the atmosphere part.

Chapter 3

The modules

3.1 Introduction

Vamps is set up is such a way that parts of the model can be used or disregarded depending on the user's needs and the available data. This implies that you don't need *all* the following information. Although the setup of the following chapters might suggest a strict separation between the modules parts overlap and parts cannot be used without some parts of other modules.

3.2 The atmosphere

In the atmosphere module, potential transpiration, potential soil evaporation and evaporation from a wet surface are calculated from meteorological and vegetation parameters. Several methods can be chosen, depending on the data available and the user's preferences. The terms potential evaporation, actual evaporation and evapotranspiration can lead to some confusion, particularly with the use of models like Vamps. Also the terms can have slightly different meanings when different methods are used. Although this is hardly desirable Vamps is no exception yet. In the final version of the model I hope to simplify all this, in the mean time most names and units remain as quoted from the original source.

3.2.1 Potential evaporation

Potential evaporation can be used in Vamps to determine actual evaporation (or evapotranspiration, depending on the method chosen) in an other part of the model. Three methods can be used:

1. Determine Penman E_0 (open water evaporation) (Penman 1956, Makkink 1957) using reflected radiation, net-radiation, relative humidity (this is used to calculate the vapour pressure deficit), wind-speed, temperature and incoming radiation.

2. Determine Penman E_0 using sun-ratio, relative humidity, wind-speed, temperature and incoming radiation.

3. Using the Makkink formula (Makkink 1961, Commissie Voor Hydrologisch Onderzoek TNO 1988) which needs incoming radiation, relative humidity, windspeed and temperature.

The input file settings for potential evaporation are described in section 6.3.6 on page 46.

Penman E_0 is determined via:

$$E_0 = \frac{\Delta R_{no} + \gamma E_a}{\Delta + \gamma} \tag{3.1}$$

Where:

R_{no}	net radiation over open water [cm/day]
E_a	aerodynamic evaporation or drying power of air [cm/day]
E_0	open water evaporation [cm/day]
Δ	slope of the saturation vapour pressure curve
γ	psychrometer 'constant' $[mbar/^oK]$[1]

The determination of E_a is described in (Calder 1990):

$$E_a = 2.6(e_s - e_a)(1 + 0.537u) \tag{3.2}$$

Where:

e_a	actual vapour pressure [mbar]
e_s	vapour pressure at saturation [mbar]
u	mean daily wind-speed at 2m

Vamps calculates e_s and e_a from relative humidity and dry bulb temperature using an equation described by Bringfelt (1986).

Net radiation over open water is given by:

$$R_{no} = R_s(1 - \alpha) - R_{nl} \tag{3.3}$$

Where:

R_s	incoming solar radiation
R_{nl}	net long-wave radiation
α	albedo of open water (0.05)

In method 1 R_{nl} is calculated using incoming and reflected short-wave radiation to calculation net long-wave radiation via:

$$R_{nl} = R_s - R_{net} - Rs_{out} \tag{3.4}$$

Where:

R_s	incoming solar radiation
Rs_{out}	reflected short-wave radiation
R_{net}	net radiation

In method 2 sun-ratio n/N is used to calculate R_{nl} via:

$$R_{nl} = \frac{86400\sigma T^4(0.56 - 0.248\sqrt{e})(0.1 + 0.9n/N)}{\lambda} \tag{3.5}$$

Where:

T	mean daily air temperature [^oK]
e	mean daily water vapour pressure [kPa]
n/N	ratio of duration of bright sunshine hours, n, to the maximum possible duration of sunshine hours, N
λ	latent heat of vaporization

Makkink (Commissie Voor Hydrologisch Onderzoek TNO 1988) reference evaporation is calculated by:

$$\lambda E = C\frac{\Delta}{\Delta + \gamma}R_s \tag{3.6}$$

Where:

C	Constant (usually 0.65)
$\lambda, E, \Delta, \gamma, R_s$	are defined as before

3.2.2 Actual evapotranspiration

If you don't want to model an entire canopy (as is described in section 3.3) you can estimate actual evapotranspiration with one the following three methods. The use of the canopy section is the preferred way.

1. Set actual evapotranspiration to be equal to potential evaporation.
2. Multiply potential evaporation by a crop factor
3. Calculate actual evaporation using the Penman-Monteith formula

3.3 Plants and trees

Although Vamps has been set up to be able to use a multilayer canopy, this option is untested and has been disabled in the alpha version. At present you are stuck to one canopy layer.

At the base of this module lies the Penman-Montheith equation(Monteith 1965).

$$\lambda ET = \frac{\Delta(R_n - G) + \rho C_p \delta e/r_a}{\Delta + \gamma(1 + r_s/r_a)} \tag{3.7}$$

Where:

ρ	density of air
c_p	specific heat of air
δe	vapour pressure deficit
γ	psychrometic constant
Δ	the slope of the saturation vapour pressure curve
r_s	canopy resistance
r_a	aerodynamic resistance

The program seperates wet and dry periods. Evaporation from the wet canopy is calculated using the Penman-Monteith equation with r_s set to zero. During these times transpiration (and thus rootextraction) is assumed zero.

The amount of net radiation available is determined via a simple fraction or via an equation by Roberts, Cabrel, Fisch, Molion, Moore and Shuttleworth (1993) linking the leaf area profile to absorbed radiation.

This module combines most of what has been explained before (in the sections concerning evaporation, transpiration and interception) and Vamps work best if you use this.

3.3.1 Interception

Introduction [2] Interception of precipitation can comprise a large part of the water balance in forested catchments. Interception is defined as (Leonard 1967):

> ...the amount of rainfall retained by the vegetation and evaporated without dripping off or running down the stem. At some point after precipitation ceases, all of the intercepted water will be evaporated...

Note that losses in the litter layer are not included[3]. The balance between precipitation, evaporation and canopy storage[4] (Rutter et al. 1971) determines the amount of rainfall interception. As the canopy parameters can change with vegetation type, so does the interception.

Interception of a forest canopy is generally determined by either the water budget method or by the aerodynamic method (Tsukamoto and Ishigaki 1989). An alternative method consists of hanging a tree in its place in the forest and weighing it continuously. This has the advantage that the evaporation during the storm can adequately be determined (Tsukamoto and Ishigaki 1989).

In tropical rain forest areas a great diversity of species exist. Highly significant differences in interception storage have been found (Leonard 1967, Herwitz 1985, Scatena 1990). This implies that parameters probably need to be determined for each site seperately.

How interception losses can be calculated is mostly determined by the available data. The model by Rutter et al. (1971) needs hourly input, while some of it's parameters should be determined using 5 minute values. In reality daily values are most commonly available, so developing equations for daily values (despite the obvious limitation) is justifiable and certainly practical (Jackson 1975). Gash (1979) developed an

[2] The settings for the input file that control interception are described in section 6.3.8 on page 46.

[3] Vamps, the final version, can estimate litter interception as a seperate term

[4] Although the word canopy implies a forest most of the methods described here can also be applied to agricultural crops or grassland

analytical variant of the Rutter model which can be used with daily input values assuming one storm a day.

Several methods for determining interception losses can be used within the Vamps model. The choice of method is mainly governed by the available input data. The following paragraphs present the methods that Vamps can use to determine interception losses.

Gash's method If input data is available on a daily basis, the method described by Gash (1979) is the preferred method. Gash's model is based on a numerical model developed by Rutter et al. (1971). In contrast to Rutter's model which is described later, it can be applied to daily measurements assuming one storm a day. Gash, Wright and Lloyd (1980) have tested both models on the same data set and found comparable results.

In the model P' – the amount of water needed to completely saturate the canopy (Gash 1979) – is defined as:

$$P' = \frac{-\overline{R}S}{\overline{E}} ln \left[1 - \frac{\overline{E}}{\overline{R}} (1 - p - p_{Tr})^{-1} \right] \tag{3.8}$$

Interception losses for storms larger and smaller than P' are dealt with separately. For small storms the equation becomes:

$$E_i = (1 - p - p_{Tr})P \tag{3.9}$$

while or large storms it becomes:

$$E_i = (1 - p - p_{Tr})P' + (\overline{E}/\overline{R})(P - P') \tag{3.10}$$

Where:

$\overline{R}$	average precipitation [mm/d]
$\overline{E}$	average evaporation during storms [mm/d]
S	storage capacity of the canopy [mm]
p	free throughfall coefficient
p_{Tr}	stemflow coefficient
P	precipitation [mm/d]
E_i	interception loss [mm]

The interception losses from the stem are calculated for days that $P \geq S_{Tr}/p_{Tr}$. As this is generally small (about 2% of gross P) it is often neglected[5]. Vamps can calculate The ratio of $\overline{E}/\overline{R}$ from the Penman-Monteith equation or use a fixed value.[6] Vamps also allows the parameters S, p and p_{Tr} to be linked to the Leaf Area Index (LAI). If lai is defined in the `interception` section of the input file, or in the `ts`

[5]This is also true for Vamps

[6]The latter is generally preferable as the penman-Monteith equation (in this case with R_s set to zero) in not very good at predicting in storm evaporation. This seems to be due to the influence of advective energy during a storm. Usually $\overline{E}/\overline{R}$ is determined by a regression of interception losses versus gross precipitation including only single storm events.

section of the input file a method described in van Dijk (1996) is used to link these parameters to LAI.

A version of the Gash model adapted to small time-steps can also be used in Vamps. In this version the canopy storage (S) is diminished with the value of the actual canopy storage at that moment. This new value is now used to calculate P' and interception losses. To use this method Evaporation during that time-step is needed as well as this is subtracted from the amount of water in the canopy at the begin of the time-step. Results of applying this adapted version to a test case using the normal version with daily time-steps and the adapted version with half-hourly data, show that interception losses as modeled with both methods did not differ more than 2%.

Calder regression model Calder (1986) presented a simple regression model for interception loss. Despite it's simplicity it could adequately describe the interception of forests in the UK. It is defined as:

$$E_i = \gamma \left[1 - exp(-\delta P)\right] \tag{3.11}$$

Where:

E_i	Interception loss [mm/d]
P	precipitation [mm/d]
γ	maximum interception loss per day [mm]
δ	fitting parameter

Interception as a fraction of LAI In this method interception is calculated as:

$$E_i = min(P, laifrac * LAI) \tag{3.12}$$

Where:

E_i	Interception loss [mm/d]
P	precipitation [mm/d]
$laifrac$	fraction (usually around canopy storage)
LAI	leaf area index

This method can describe total interception rather well but fails on a storm to storm basis (it is used in the TOPOG model (Vertessy et al. 1993)). The fact that it is based on LAI makes it well suited for applications in which the canopy changes in time, although an alternative Gash model in which the parameters S, p_{tr} and p are related to LAI might yield better results (see also paragraph 3.3.1 on page 27).

Rutter's model Rutter et al. (1971) (see also Rutter, Morton and Robins (1975)) developed a model of interception based on the storage of water in the canopy. The model needs hourly meteorological data plus canopy parameters as input, although five minute data are preferred.

$$dC/dt = P(1 - p - p_{Tr}) - (C/S)E_c - D_r \tag{3.13}$$

Where:

t	time-step [hour]
C	amount of water in the canopy
P	gross precipitation [mm/h]
p	free throughfall coefficient [-]
p_{Tr}	proportion of water discharged via stem flow
S	canopy storage capacity [mm]
E_c	evaporation rate from wet canopy [mm/h]
D_r	drainage-speed from storage in canopy [mm/h]

The model is rather insensitive to both S and p (Gash and Morton 1978).

3.4 The soil

3.4.1 Introduction

The soil module can be regarded as the core of Vamps. The model was developed using the SWAP94 model as a framework. About 80% of Vamps's soil module consists of hand-translated[7] Fortran code from the original SWAP94 program. The head based solution of Richard's equation using remains more or less the same in the present version, although the solution has been extended with a band-diagonal procedure and a (optional) step to regain full machine precision (described in Press, Teukolsky, Vetterling and Flannery (1992)) that can be used if a zero pivot occurs in the tri-diagonal solution.

3.4.2 Solving Richard's equation

The soil module solves Richards equation using a ψ based form of this equation. A differential moisture capacity ($C(h)$) has been introduced to have a single independent variable. This differential soil moisture capacity is equal to the slope of the soil-moisture retension curve, $d\theta/d\psi$.

$$\frac{\partial \psi}{\partial t} = \frac{1}{C(\psi)} \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} - 1 \right) \right] \tag{3.14}$$

A sink term (S) has been added to accommodate water extraction by roots and lateral flow.

$$\frac{\partial \psi}{\partial t} = \frac{1}{C(\psi)} \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} - 1 \right) \right] - \frac{S(\psi)}{C(\psi)} \tag{3.15}$$

Where:

[7] And changed to my liking

ψ	Suction head [cm]
t	time [days]
C	differential moisture capacity ($d\theta/d\psi$)
S	sink term
z	heigth [cm]

At present the reader is referred to Feddes et al. (1978) and Belmans et al. (1983) for a detailed description.

Vamps first estimates the largest time-step possible with the code presented in example 3.4.1. It uses the previous dt and a value for thetol (the maximum change in theta allowed per time-step) to do so. This is done for all layers in the soil profile. This way the layer in which the largest changes occur determines the maximum dt. If this produces results that are regarded as not accurate enough (set with the `thetol` variable) Vamps will perform iterations to get the accuracy asked. If this fails dt is halved and the procedure is started again until the accuracy is reached or dt-minimum is reached. The maximum number of iterations is set with `maxitr` variable while the minimum allowed dt is set using the `dtmin` variable.

3.4.3 Relation between Theta, ψ and K_{unsat}

Vamps was first developed using the equations by van Genuchten (1980) (see also Mualem (1976)). This is still the default method. For extra speed and flexibility Vamps can also read tables which describe these relations. This means that the user can use any method available[8]. At present Vamps can only generate tables for the van Genuchten method. However, it can read tables generated by the `topog_soil` program (Beverly 1994) or user made tables, thus extending the amount of models for the theta vs ψ that can be used to anything the user can imagine. The following is derived from the topog_soil[9] manual:

> Six different soil tables can be generated using topog_soil, the options being:
>
> 1. Broadbridge and White (1988)
> 2. Campbell (Ross and Bristow, 1990)
> 3. Campbell (SWIM model version, 1974;1985)
> 4. Campbell (Ross, Williams and Bristow, 1991)
> 5. van Genuchten (1980)
> 6. Simplified Bucket Model soil table (1979)

At present only items 1 and 5 have been tested using Vamps. A description of the file format for these tables can be found in Appendix ??.

[8] the use of lookup tables can in certain circumstances also cause some speed increase

[9] A version of topog_soil for the Vamps model will be available in the future

Example 3.4.1 The C code used by Vamps to estimate the maximum dt

```
/*-
 * void timestep(int step)
 * Calculation of timestep (dt) depending on theta  changes
 * and new time (t)
 */
double
timestep (int step)
{
  int i;

  /* Store previous values */
  tm1 = t;
  dtm1 = dt;

  /* start with max dt */
  dt = dtmax;

  mdtheta = fabs (thetm1[0] - theta[0]);
  for (i = 0; i < layers; i++)
    {
      dtheta = fabs (thetm1[i] - theta[i]);
      mdtheta = mdtheta < dtheta ? dtheta : mdtheta;
    }

  if (mdtheta < 1.0E-6)
    dt = dtmax;
  else
      dt = 10.0 * thetol * dtm1 / mdtheta;

  /* restrict to dtmin and dtmax */
  dt = dt > dtmin ? dt : dtmin;
  dt = dt < dtmax ? dt : dtmax;

return dt;
}
```

Van Genuchten Vamps default and most tested method of describing the theta versus suction- head and k_{unsat} is base in the van Genuchten equations (Mualem 1976, van Genuchten 1980). The dimensionless water content is given as:

$$\Theta = \frac{\theta - \theta_r}{\theta_s - \theta_r} \tag{3.16}$$

Where:

θ_s	saturated water content
θ_r	residual water content
θ	actual soil water content
Θ	dimensionless soil water content

Then a class of $\Theta(\psi)$ functions can be given by:

$$\Theta = \left[\frac{1}{1 + (\alpha\psi)^n}\right]^m \tag{3.17}$$

Where:

ψ	pressure head
α, n and m	soil characteristic parameters

if $m = 1 - n/m$ the following closed-for expression of $K(\Theta)$ can be obtained:

$$\Theta = \begin{cases} \frac{1}{[1+(\alpha\psi)^n]^{1-1/n}} & \psi < 0 \\ 1 & \psi \geq 0 \end{cases} \tag{3.18}$$

$$K = K_s\Theta^{1/2}\left[1 - \left(1 - \Theta^{n/(n-1)}\right)^{1-1/n}\right]^2 \tag{3.19}$$

The $K_s\Theta^{1/2}$ part of the equation can be changed to $K_s\Theta^L$ is which L is a fitting parameter describing the Θ to K_{unsat} relation if you have K_{unsat} measurements.

Values for the parameters α, n and L for most soil types in the Netherlands can be found in Wösten, Bannink and Beuving (1987). Such values for tropical soils seem to be less common. In that case a nonlinear regression method can be used to determine the parameters α and n from a soil moisture retention curve (Marquardt 1963).

Clapp/Hornberger A power curve can conveniently describe the moisture characteristics of a soil as has been demonstrated by **?**. Clapp and Hornberger (1978) described these functions and presented some representative values for the soil hydraulic parameters needed. They are shown in Table 3.4.3.

$$\psi = \psi_s\left(\frac{\theta}{\theta_s}\right)^{-b} \tag{3.20}$$

Soil hydraulic conductivity functions are derived by integrating a function of ψ with respect to θ. This results in the following equation:

$$\frac{K}{K_s} = \left(\frac{\theta}{\theta_s}\right)^{2b+2} \tag{3.21}$$

Soil Texture	$\overline{b}$	$\overline{\psi_s}$	$\overline{\theta_s}$	$\overline{K_s}$
Sand	4.05	12.2	0.395	1520.64
Loamy sand	4.38	9.0	0.410	1350.72
Sandy loam	4.90	21.8	0.435	299.52
Silt loam	5.3	78.6	0.485	62.21
Loam	5.39	47.8	0.451	60.05
Sandy clay loam	7.12	29.9	0.420	54.43
Silty clay loam	7.75	35.6	0.477	14.69
Clay loam	8.52	63.0	0.476	21.17
Sandy clay	10.4	15.3	0.426	18.72
Silty clay	10.4	49.0	0.492	8.93
Clay	11.4	40.5	0.482	11.09

Table 3.1: Representative values for hydraulic parameters. After Clapp (1978).

Where:

ψ	suction head
ψ_s	suction head at saturation (air entry value)
θ	volumetric water content
θ_s	water content at saturation (porosity)
b	soil characteristic parameter
K	hydraulic cnductivity
K_s	saturated hydraulic conductivity

An input file containing all the soil-types mentioned in table 3.4.3 can be found in the src/sllib directory. It is called `soillib.inp`.

Brooks and Corey **?** proposed a power-law description of the soil moisture characteristic that uses an explicit air-entry (bubbling) pressure head h_b below which the soil begins to desaturate. The effective saturation S_e is defined as:

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \tag{3.22}$$

Where:

θ_s	saturated water content
θ_r	residual water content
θ	actual soil water content

The retention curve and unsaturated conductivity are then given by:

$$K = K_s \, S_e^{2/\lambda+3} \tag{3.23}$$

Where:

h	pressure head (negative for unsaturated conditions)
h_b	air-entry (bubbling) pressure head ($h_b < 0$)
λ	pore-size distribution index
K_s	saturated hydraulic conductivity

The conductivity can equivalently be written as a function of pressure head:

$$K(h) = K_s \left(\frac{h_b}{h}\right)^n, \quad n = 2 + 3\lambda \tag{3.24}$$

Starting approximations for converting van Genuchten parameters to Brooks-Corey are $\lambda \approx n_{\mathrm{vG}} - 1$ and $h_b \approx -1/\alpha$. The two retention curves agree closely for $S_e \leq 0.80$; the main difference is near saturation where Brooks-Corey has a discrete air-entry while van Genuchten desaturates smoothly.

3.4.4 Bottom boundary conditions

Vamps can handle several boundary conditions at the bottom of the profile. Below is a lists of those available in the model. Not all conditions have been fully tested in the present version of Vamps.

- A no-flow bottom boundary (5)
- Free drainage at the bottom of the profile (6)
- A fixed head at the bottom of the profile (4)
- A fixed flux at the bottom of the profile (1)
- Daily groundwater depth (0)
- Seepage or infiltration from/to deep groundwater (2)
- Flux calculated as a function of head (3)

The choice of bottom boundary condition is of great importance to the results of the modelling. Unfortunately most of the time the choice of bottom boundary is usually determined by the available data or lack thereof.

See **vamps**(5) for details[10].

[10]The numbers between brackets denote the value of the `bottom` variable in the `soil` section needed

Part II

User’s guide

Chapter 4

Installation of Vamps

Because Vamps runs on several quite different platforms installation requires some things to be set. Vamps is distributed as source code only or as source code plus binaries. As I only have access to MS-DOS, Linux, Sun-OS and OS/2 systems these are the only binaries I can make.

In the case of a UNIX system installation from source code the following steps are required:

1. First read the files `0readme` and `0install`.

2. Run the `configure` script in the `src` directory. If you don't want to install Vamps in `/usr/local` make sure to run the `configure` script with the `--prefix=/your/installdir` option. If you have the agl or **S-Lang** library installed in a non-standard place set the LDFLAGS environment variable so that `configure` can find them (e.g. `setenv LDFLAGS -L/home/schj/lib`). If you don't have **S-Lang** now would be a good time to install it.

3. Check the top part of the Makefile in the SRC directory

4. Run make lib, make all and make install.

In case of an MS-DOS system (it should be a 386Sx at least) it is sufficient to copy all the .exe files to a directory in your search-path. You should also copy the default `vamps.rc` and `vamps.sl` files – all located in the sllib directory – to a place where Vamps can find them (set the `HOME` or `VAMPSLIB` environment variable to the directory in which these files reside).

You will need to have the **S-Lang** library installed if you want to compile the program from source. The **S-Lang** library can be found at `ftp://space.mit.edu/pub/davis/slang`.

Chapter 5

Program organization

5.1 Introduction

The main module of Vamps is the Vamps program itself. It uses a series of equations to describe the flow of water through a forested ecosystem. Preparing input for the Vamps program takes quite a while and is by no means a trivial task. Some 'basic' understanding of the functioning of forested ecosystems is needed.

REWRITE THIS !!!

To facilitate the data pre- and post-processing some tools are included: The **wrini**(1) program can be used to create input files at a basic level. It is mostly used in scripts. The **issec**(1) and **getstr**(1) programs can be used to extract part of output files. For general post-processing however, the **vsel**(1) program is recommend. This program can produce data the can be fed directly to plotting programs such as **gnuplot**(1) and **graph**(1). **vsel** output can also be imported into popular spreadsheet programs. [1]

Vamps can crash in certain situations. This can have three reasons: *(1)* I have made an error, causing you to discover a genuine bug, (2) the situation you want to model is beyond the capabilities of Vamps, *(3)* the timestep estimate done by Vamps is not right. The method that is used to solve Richards equation is rather stable but can go bezerk given the right combination of boundary conditions and soil parameters. A lot of precipitation on a *very* clayey soil is a recipe for disaster. There are some options you can use to circumvent some problems. They are described in section 7.4

5.2 System requirements

If you are not in a hurry a 386Sx MS-DOS based computer with a 1.44Mb floppy drive is the minimum you need for running Vamps. As most calculations are done with floating point numbers, a 387 co-processor will speed up things a lot[2]. Djgpp[3]

[1] Using **S-Lang** it should be fairly easy to write custom postprocess-functions. For example the following command gets the variable `volact` from an output file for all timesteps and write a two column ascii file that can be read into spreadsheet software: `mwrite(getts("volact","vamps_output_file",-1),"columnfile");`

[2] Dramatically might be a better word as I found out myself

[3] A MS-DOS port of the GNU C/C++ compiler which I used to create the .EXE files.

compiled programs require a DPMI server[4] and should run with 512 Kb of memory, although 1Mb should be considered a minimum. You need some disk space as well. The Vamps binary does not use a lot of it, but your input (and certainly output) data will probably by the biggest users of disk space. Vamps is known to run with several operating systems: MS-DOS, Linux, Net-BSD, Sun-OS, AIX 3.2 and OS/2 Warp.

5.3 Defaults file

REWRITE!! Several layers!

Some settings which you want to use in several sessions can be added to the Vamps defaults file. The name of this file is `.vampsrc` on unix systems or `vamps.rc` or FAT based (MS-DOS or OS/2) systems. Vamps first searches for this file in the current directory, if this fails the directory to which the HOME environment variable is set is searched. After this the directory to which the VAMPSLIB environment variable is set is searched. If this also fails the compiled-in `vampslib` directory is searched. If no defaults file is found compiled-in defaults will be used. The syntax of the defaults file is the same as the general input file. Allowed sections and variables are described in section 6.3.1 on page 44.

If Vamps is compiled with **S-Lang** support it also needs to load several **S-Lang** files at startup. These must be present in the VAMPSLIB directory. If (this is an MS-DOS example) you have installed Vamps in c:\vamps you should use the following command to let Vamps find it's files:

```
set VAMPSLIB=c:\vamps
```

In UNIX the command to set an environment variable differs per shell used, in the c-shell you could use:

```
setenv VAMPSLIB /home/usr-x/lib/vamps
```

5.4 Time series and time steps

Time series are values of certain variables over time that Vamps uses as input. An example of a time series is precipitation. All time series [5] must be listed in the ts section of the input file.

The precipitation time series is quite special because it *must* be present. This is because this time series determines the time steps that Vamps make with it's x-column. This column should be in days or fractions of day (i.e. 1.5 is halfway day 1). Time steps may vary within one run. The following example of a precipitation input file shows how this is done:

```
# This is a comment
# Precipitation from the no-name site
```

[4] A freeware DPMI server is included in the distribution. Note that a windows or OS/2 MS-DOS box can also provide DPMI service.

[5] Time series files must contain two columns, time and value. Lines starting with # are regarded as comments (see **ts(5)**).

```
# time in days         amounts in cm/day (intensity!)
0.5  0.6
1.0  0.9
2.0  0.7
2.1  0.9
2.2  0.7
2.3  0.5
3.0  0.8
3.5  0.0
```

In this case 0.9 cm/day fell between day 2.0 and day 2.1 amounting to $0.9(2.1 - 2.0 = 0.09)$ cm. There are several things that must be remembered with respect to time-steps and input-files in Vamps:

1. Units are in cm/day. With precipitation you might have to convert your data to intensities.

2. The time in the x-column is the *end* of an interval. In of the above example output given at time 3.0 is calculated from the situation at time 2.3 after including the precipitation from time 3.0. The duration of this time-step is 0.7 days (3.0 - 2.3). This implies that in the case of the first record no duration of the time-step is known. Vamps defaults to one day for the first time-step. You can change this using the `firststep` variable in the `time` section of the input file.

3. At present the x-column in all files but the precipitation-file is ignored (that is, it must be present, but Vamps does not check the time-values).

5.5 The principle of layers in the soil section

The `layers` variable in the `soil` section determines the number of layers used by Vamps in the *calculations* [6]. The number of *physical* layers (with different K_{sat} for example) is determined by the number of `layer_n` sections you have. Figure 5.1 shows this principle. The users specifies the first layer of each physical layer. Any layer that is not specified automatically inherits the settings from the overlying layer.

[6] Actually **nodes** would be a better name for this variable

The layers variable in the soil section of vamps specifies the number of layers used in the calculations, NOT the amount of physical layers. This figure displays the relation between the layers used in the calculations and the physical layers. You only need to specify the characteristics of the top layer of each block of layers with identical physical properties.

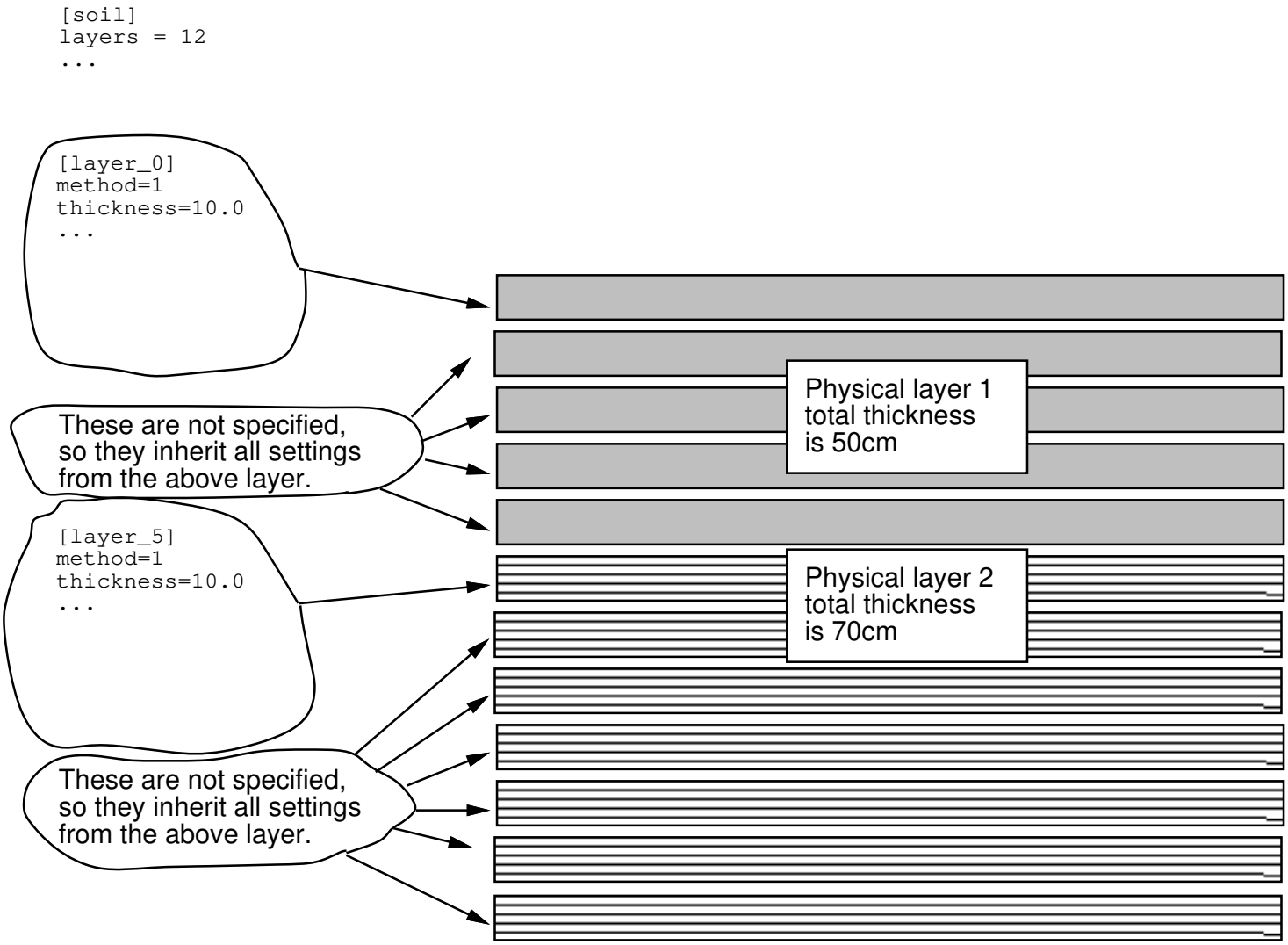


Figure 5.1: Relation between the layers variable in the soil section and the amount of physical layers.

Chapter 6

The input file

6.1 introduction

Vamps reads it's options and most of the data needed for modelling from the input file. This file is in plain ascii format and is divided in several sections. The exact format of this file is described in section 6.2. An input file must be created before running Vamps. There are two way of creating an input file for Vamps. *(1)* Use the (not finished) Vamps preprocessor. This preprocessor[1] will let you choose from menu items to select all the options that are set in the input file and generate an input file for you once you are finished. Or you can *(2)* use an ascii editor or a word-processing package to create this file.

6.2 Input file format

The input file format for Vamps is much like Microsoft Windows `.ini` files. All names and sections are case insensitive. The case of string variables is unaltered so filenames are case-sensitive if the operating system is. The file is divided into sections which each contain variables that have some connection with each other. Each variable name is followed by an equal sign (=), after which the value of the variable is placed. Example:

```
[vamps]
verbose = TRUE
# This is a comment

[environment]
caseid = Interception test file. Bisley catchment.\
Summer of 1995
```

The \ character may be use to break up long strings over more than one line[2]. The # sign at the start of a line is used to denote a comment Only the first '=' sign is significant. Spaces within the names are copied verbatim, and thus become part of the name. This makes the following construction legal:

[1] The use of this preprocessor is described in a seperate manual

[2] The character itself is converted to a space

```
[vamps]
output mode = This is ('=') a nonsense\
example.
```

In this case the variable 'name' in section 'vamps' is 'output mode' and it's value is 'This is ('=') a nonsense example'.

[3].

Vamps uses six types of variables: *(1)* floats (floating point numbers), *(2)* arrays (a row of floating point numbers separated by whitespace), *(3)* strings (a series of characters), *(4)* integers (whole signed numbers), *(5)* characters (a single printable character) and *(6)* boolean (special kind of integer, either 0 or 1 – FALSE of TRUE, NO or YES).

6.3 Available sections and variables

All sections with available variables and their defaults values as well as the type of the variable are listed below. The most up to date list is presently found in the **vamps(5)** manual page. You will need this list badly when you construct an input file for VampsÒnce the graphical preprocessor is finished things will become a little easier.

Vamps has been developed using the **swap94** FORTRAN code as a starting point. Most names of variables have been changed (sorry) but some of the variable names as well as their physical meaning have not been altered.

6.3.1 [vamps]

verbose If set to FALSE Vamps will be silent, if set to TRUE Vamps will display progress information

header [4] Is set to FALSE no header in will be added to the output, if set to TRUE an header will be added to the output. Defaults is FALSE

logging FALSE = no logging is done, TRUE = logging is on

logfilename name of the file to which logging is performed

iniinmem TRUE = the input file is read into memory (some speedup), FALSE = the input file is not read into memory

progstr String which is used to display progress information (use 0, 1 or 2 for build in strings). 0 will show calculation time and estimated time to go. 1 will show a percentage finished bar and 2 will show 'calculating'.

You can define your own such as:

```
Vamps is running, please
wait.....................
```

[3] If you want you can change this character using the `commentchar` variable in the `vamps` section or the -C command-line option. Usually there should be no reason to do so

[4] See also the - -Header option in Vamps (1)

graphcommand full path to `gnuplot` with optional command line options. In OS/2 only gnuplot.exe should be specified here. This seems to be a 'bug' in the emx library which I used to make the binaries.

commentchars Character(s) that denotes the start of a comment. This defaults to #% and the first character should *not* be changed unless you have a very good reason to do so.

xtrasl comma seperated list of **S-Lang** files to be loaded a startup

6.3.2 [time]

Vamps will not run without a time section in the input file with at least the `steps` variable set[5].

steps Integer value specifying the number of steps in the current simulation. This value should be smaller or equal to the number of entries in the precipitation file.

startpos Position (line) in the precipitation input file used as start. Counting starts at zero.

starttime day at which the simulation should start If this value is not specified simulation starts at the first step in the precipitation input file. If both starttime and startpos are specified starttime will be used.

6.3.3 [run]

outputfile Filename to save output to. You can override this with the o command line option.

runid No used at moment

6.3.4 [xout]

filename Filename for extra output in column type format. No extra output is generated if this variable is not present.

6.3.5 [determine]

canopy determine canopy (see section canopy)

evaporation determine actual evaporation (see section evaporation)

pevaporation determine potential evaporation (see section pevaporation)

soilmoisture determine soilmoisture profile (see section soil)

fit Use non-linear regression to try and fit to measured data (see fit section)

[5] This will change in the beta version in which time-steps will be determined by the time column in the input file

6.3.6 [pevaporation]

method 0 = potential evaporation via Penman E0 (Need: refrad, netrad, rhumid, windspeed, temp, inrad), 1 = potential evaporation via Penman E0 (using sunratio) (Need: sunratio, rhumid, windspeed, temp, inrad), 2 and 3 = not yet done, 4 = potential evaporation using Makkink (Need: rhumid, windspeed, temp, inrad)

6.3.7 [evaporation]

method 0 = evaporation equal to potential evaporation, 1 = multiply potential evaporation by a crop factor (need cropfac), 2 = calculate actual evaporation using the Penman-Montheith formula

cropfac A floating point number representing the crop factor with which the potential evaporation is to be multiplied to yield actual evaporation.

6.3.8 [interception]

method 0, 1 , 2, 3. gash, rutter laifrac or calder

gamma gamma in calder equation

delta delta in calder equation

E_avg/R Evaporation/Average Rainfall during a storm in gash. If this is not set penman/Montheith will be used with R_a set to zero.

p_tr Fraction of water diverted to the trunk (gash, rutter)

p_f Free throughfall coefficient (gash, rutter)

S Canopy storage in cm (gash, rutter)

gashm Either 1 or 0. If set to 1 an adapted version of gash is used. This version should work for time-steps smaller then 1 day. Default = 0

laifrac the canopy interception coefficient

lai The canopy leaf area index. This is needed for the laifrac method. If it is not present it is searched in the canopy section.

6.3.9 [canopy]

Although you don't have to use the canopy module it is recommend that you do so if possible. The other methods of determining transpiration, interception etc. don't have a close interaction with the soil modules and usually provide poorer results.

layers Number of canopy layers (this is largely determined by the accuracy of your LAI profile). At the moment only one layer is allowed.

Rnet_absorb The fraction of the total radiation absorbed by the canopy ($0 \leq Rnet_absorb \leq 1$). The remaining amount will be used for soil evaporation.

transpiration Which transpiration equation should be used.
2 = penman Montheith

z Height of the canopy (m)

z_0 Aerodynamic roughness length (m)

d Zero plane displacement length (m)

rs Canopy resistance (s/m). If this is not specified the user defined regression equation **estrs()** will be used.

drytime If this variable is set this value (in days) will be used to determine how long it takes for the canopy to dry.

wetevap If this variable is set this value (in cm/day) will be used to determine the canopy wat evaporation rate in stead of Penman-Montheith with Rs set to zero.

6.3.10 [roots]

depth depth of the root zone in cm. If you want rooting depth to change in time you should use the `drootz` variable in the `ts` section.

swsink 0 = sink term according to Feddes. (Need: hlim1 hlim2u hlim2l hlim3h hlim3l hlim4)
1 = sink term according to Hoogland. (Need: hlim1 hlim2u hlim2l hlim3 hlim4)

swhypr 0 = linear relation between the points hlim3 and hlim4 of the sink term.
1 = hyperbolic relation between the points hlim3 and hlim4 of the sink term.

swupfu 0 = water uptake function according to Feddes.
1 = water uptake function according to Hoogland.
2 = water uptake function according to Prasad (1988).

cofsza Intercept a in Feddes et. al 1988 (only needed if swupfu =1).

cofszb Slope b in Feddes et. al. 1988. (only needed if .B swupfu =1).

hlim1 Pressure head value (cm) below which roots start to extract water from the upper soil layer (starting point).

hlim2u Pressure head value (cm) below which roots start to extract water optimally from the upper soil layer.

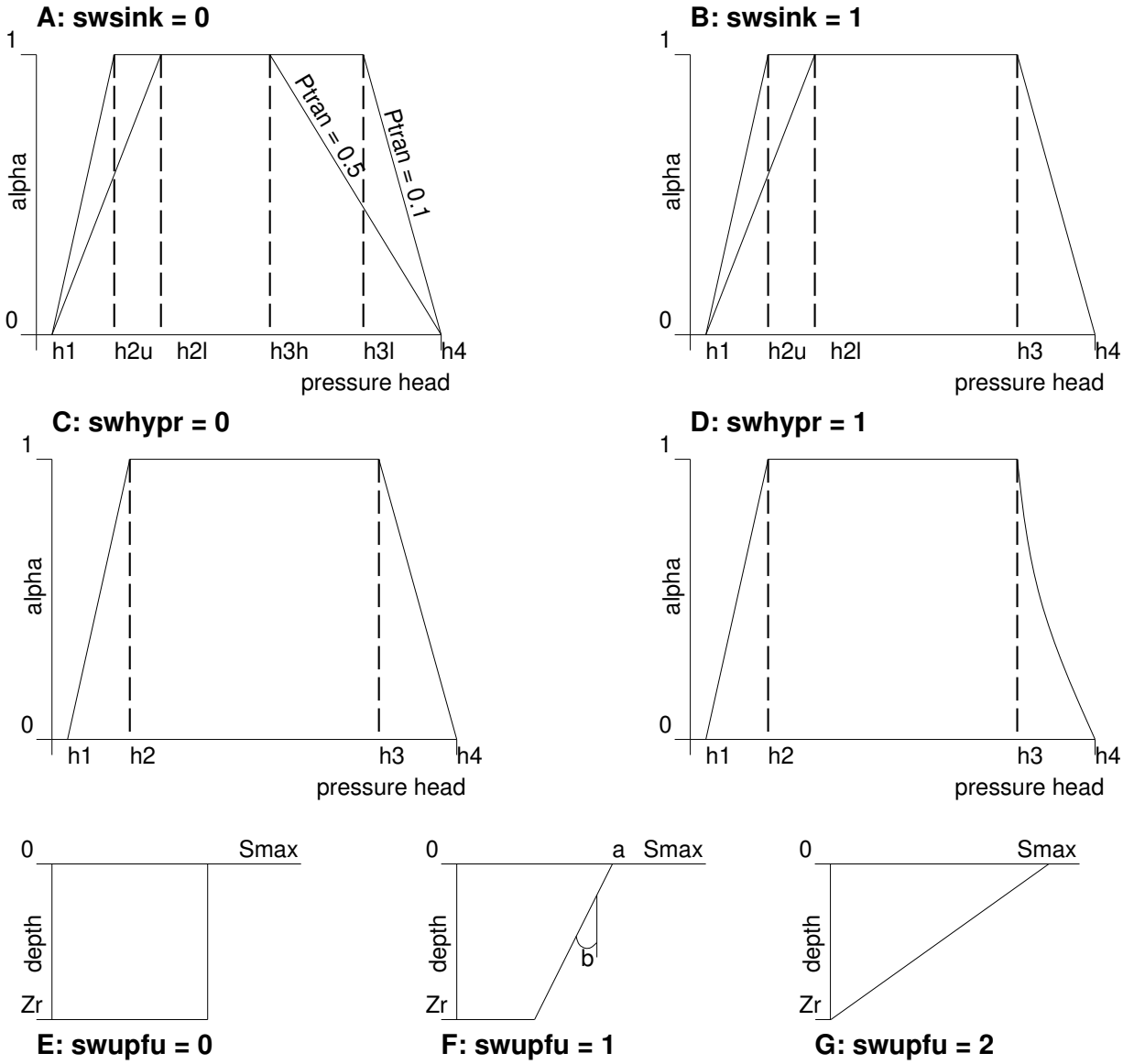
hlim2l As above, but for all lower soil layers.

hlim3h Pressure head value (cm) below which roots cannot extract water optimally any more, for a High pot. transpiration rate equal to 0.5 cm/d (limiting point).

hlim3l As above, but for low pot. transpiration rate equal to 0.1 cm/d.

hlim3 Pressure head value (cm) below which roots cannot extract water any more (limiting point).

hlim4 Pressure head value (cm) below which no water uptake by roots is possible (wilting point).



6.3.11 [ts]

The ts section has one entry that *must* be present: precipitation. The precipitation file is important because it also determines the time-steps at which output is calculated. More information about time-steps can be found in section 5.4.

The file format consists of colums separeted by whitespace (space or tabs). Vamps assumes the first column to hold the time and the second column to hold the value. You can override these assumptions by appending `,xcol,ycol` to the filename[6]. Counting begins at zero.

The ts section can be handy if you don't like the method(s) which Vamps can use for determining things like potential evaporation. Simply put them in here and Vamps will use the values you supplied instead.

[6]This implies that a comma is *not* allowed in a filename

pevaporation Name of file with potential evaporation data (if not calculated)

ptranspiration Name of file with potential transpiration data (if not calculated)

interception Name of the file with interception data (if not calculated)

precipitation Name of file with precipitation data. This file file also determines the time-steps at which output is calculated. See section 5.4 for more information.

drootz Name of file with rooting depth (cm) in time. You need to specify at least three points. Other points will be interpolated using a spline if they don't exist. See the file `src/tc/ts_spl.c` for the interpolation routine.

6.3.12 [soil]

outskip Skip outskip timesteps in soil output. Used to reduce outputfile size.

initprof 0 = water content profile (need theta_initial array), 1 = pressure head profile (need h_initial array) and 2 = Calculate pressure head profile (need gw_initial in soil section).

gw_initial initial ground-water level in cm below field-level (needed if initprof = 2)

swredu Reduction of soil evaporation
0 = no reduction, 1 = the Black (1969) model is used, 2 = the Boesten (1986) model is used and 3 = an adapted version of the Boesten (1986) model is used. This version takes into account the actual moisture condition of the soil surface.

cofred The factor alfa in Black or Beta in Boesten. This is not needed for swredu = 0.

bottom Bottom boundary condition: 0 = daily ground-water table depth (cm) is input, 1 = Given flux, 2 = seepage or infiltration from/to deep ground-water, 3 = Flux calculated as a function of h, 4 = interpolation between daily values of given pressure head, 5 = Zero flux at bottom, 6 = Free drainage.

smooth Integer value giving the size of the running average used for smoothing the ksat, theta_saturation and residual_water profile. Set to zero for no smoothing (default).

layers] Number of soil layers in calculation (how many real (physical) layers exist is specified by the amount of layers_*n* sections[7])

pondmx Maximum amount of ponding (in cm) allowed at the top of the profile, defaults to 0.0.

gwlevel Water level at bottom of profile, needed if bottom = 4

dtmax Maximum time-step (in days) in soil module

[7] see also section 5.5 on page 41

dtmin Minimum time-step (in days) in soil module. Both dtmax and dtmin should probably be left alone. By setting dtmin and dtmax to the same value Vamps can be forced to use a fixed time-step.

speed Integer value ranging from 1 (slow) to 6 (fastest) determining the tradeoff between calculation accuracy and speed. This options combines the settings in dtmin, thetol, solvemet, mktable, maxitr and swnums.

If you also specify one of these variables seperately the settings from speed are overridden. Default for speed is 3.

maxitr Maximum number of iterations in the soil module. Iterations are only performed if swnums == 1.

thetol Theta tolerance. If swnums != 1 (by default) Vamps will use this value to check if the solution is good enough and perform iterations if needed. Setting this value to high or low gives poor results. The default value usually works fine. If you have no problems you better leave this one alone. Usually this variable should be between 1.0E-2 and 1.0E-5.

solvemet Determines how vamps solves the equation matrix. If set to 0 the default (tridiagonal) solution is used. If set to 1 the soil module will always treat the solution as a band-diagonal matrix. If set to 2 a very general solution is used. This solution includes a step to regain full machine precision but is rather slow.

swnums If set to 1 the soil module won't check for convergence of solution. It assumes that the initial maximum dt is a good guess. Things can be quit a lot faster in all cases but can give poor results in most cases as well.

mktable if set to true Vamps will create look up tables for the theta vs K relation, and use those in stead of the function during iteration. By default this option is set to false. Set to true to speed up calculation. The solution can become instable at high suction heads. In stead of letting Vamps generate the look-up tables you can also read pre-made tables using the method option in the layer_n section.

Depending on the problem, the amount of memory you have installed, the floating-point performance of you CPU and the optimizations your compiler can make, this can speed up calculations by 50%. You will lose some precision in the process. By default the program uses 300 points long look-up tables. Use the `tablesize` variable in the `soil` section to change this value.

There is no exact way to determine the speed increase you will get. For example, on a 66MHz 486DX2 running Linux, a 25% speed increase was established without compiler optimization (gcc) while a 50% speed increase was measured when de program was compiled with optimization. It seems that gcc does a better job optimizing the look up procedure than the intensive floating point calculations in the van Genuchten equation[8]. On a RISC processor with better floating point performance this balance may be different, and you probably won't get

[8] actually the all important thing here is how fast your compiler/CPU combo can execute the pow() function as this usually account for 90% of the CPU time!

speed increases of 50%. On a rs6000 365 running AIX 3.2 the mktables option resulted in a 23% speed increase.

estdmc if set to true and the `mktable` variable is also true the dmc table will be made using ts_slopes and the Pf curve in stead of the `h2dmc` function. If you use **S-Lang** soil functions and you define `estdmc` to be true you don't need the `h2dmc` function.

tablesize Sets the size of the look-up tables. Defaults to 300. Increase this for better accuracy at a penalty of using more memory and some performance loss. Provided the program fits in physical memory table sizes up to 1200 (very large) points still give some speed improvements compared to not using the look-up tables. Sometimes decreasing this to 80 or so can be done without much loss of accuracy or stability.

dumptables If set to true the look up tables will be dumped to the initial section of the output-file; x and y in separate array type variables for each layer. This can be handy for plotting pF curves for example.

verbose Verbose level in soil module (0 = silent). I recommend a setting of 1 (default) which will show you how long Vamps will run.

smddepth If this variable is set the SMD (soil moisture deficit) will be calculated until this depth. Otherwise the rooting depth will be taken.

fieldcap Head in cm at which the soil is at field capacity. Needed for determination of soil moisture deficit. Default = -100.0

6.3.13 [drainage]

This is presently in a very experimental stage. One day when Vamps will be two-dimensional this will become more important.

method Variable which controls the type of lateral drainage.

Value	Description
0	No lateral drainage (default)
1	TOPOG type drainage (only at saturation) and
2	Allow also unsaturated lateral flow

If the `drainage` variable is larger than zero the `slope` variable must be set also.

slope Slope used in the calculation of lateral drainage (floating point value).

exclude Array with layers in which lateral drainage is not allowed. 0 ¡= value ¡ layers example:

```
exclude = 12 1 23 45
```

You can use a construction like this and a no-flow bottom boundary to simulate a lysimeter.

6.3.14 [*soilsectionname*]

This section can have *any* name and contains the soil specific info. The `soilsection` variable in the `layer_n` section refers to the name of this section.

method Method for k vs theta relation:

Value	Description
0	Clapp/Hornberger
1	van Genuchten
2	not yet implemented
3	van Genuchten parameters are determined from theta vs pf pairs
4	read TOPOG _soil soil tables
5	user defined Python functions
6	Brooks and Corey (1964)

ad 3) Given values of alpha and n are used as initial guesses. Required variables: `theta` (array of values) and `pf` (array of values). Optional: `alpha` and `n`. The exponent l is set to 0.5,

ad 4) Vamps has the ability to read and use soil tables generated by the TOPOG _soil program. It will then use these look-up tables in stead of using direct calculations. In future Vamps will be distributed with it's own version of the _soil program

ad 5) In this case the user must define the following S-Lang functions:

Function	Description
`Void getspar(char *section, Int nr)`	called to get additional parameters from the input file
`Float h2dmc(Int nr, Float head)`	calculates suction head to differential moisture capacity
`Float t2k(Int nr, Float theta)`	calculates k_{unsat} from moisture content (theta)
`Float t2h(Int nr, Float theta, Float depth)`	calculates suction head from moisture content (theta)
`Float h2t(Int nr, Float head)`	calculates moisture content from suction head

If you use this option you should either have a *very* fast computer or use the `mktable` option in the `soil` section. The following tables list some example excecution time when using the some method coded in C and **S-Lang**.

Method	time [sec]
C – no tables	25.37
C – tables	18.52
SL – no tables	393.90
SL – tables	19.49

An example can be found in the file func.sl in the vampslib directory.

description An optional description of the soil layer.

ksat Saturated hydraulic conductivity of the layer.

kh/kv Ratio of K_{sat} horizontal divided by K_{sat} vertical. This is only used if you use lateral drainage (see drainage section). By default this is set to one.

thetas Theta at saturation (porosity).

psisat Head at saturation (air entry value) needed for Clapp/Hornberger

b Factor b in Clapp/Hornberger

theta_residual Residual amount of soil moisture.

alpha Alpha in van Genuchten.

l l in van Genuchten (use 0.5 if not determined). Normally (unless you have determined a theta vs K_{unsat} relation) you don't have this information.

n n in van Genuchten.

lambda Pore-size distribution index λ in Brooks and Corey (method 6). A starting approximation from van Genuchten parameters is $\lambda \approx n - 1$.

hb Air-entry (bubbling) pressure head h_b in Brooks and Corey (method 6), in cm. Must be negative (e.g. `hb = -16.4`). A starting approximation from van Genuchten parameters is $h_b \approx -1/\alpha$.

tablefile File from which the soil table should be read. Use this if method = 4.

tablefiletype Type of the table-file. All filetypes can have comment-lines starting with #.

Value	Description
1	TOPOG-type Vamps only uses columns 1, 3 4 and 5
2	White space separated columns (psi theta k)
3	White space separated columns (psi theta k diff_moist)

ad 2) In this case the differential moisture capacity (d_theta/d_phi) is estimated.

Note that all tables should be made with *descending* theta values.

6.3.15 [layer_*n*]

Only the `layer_0` section is a must. The rest is only needed if you have more than one physical soil layer.

thickness Thickness of the layer (in cm).

soilsection Name of the section that contains the soiltype info for this layer (node).

6.3.16 [fit]

fit Data-set used in fitting

fitto Name of a time serie (must be specified in the ts section) with actual measurements

n (n = 0 ... n= MAXPAR) Add a new parameter to the fitting list. At present the following parameters are available:
1 (n)
2 (alpha)
3 (ksat)
4 (b)
5 (psisat)
6 (l)
7 (residual_water)
8 (thetas)

layer Physical layer to use in fitting.

6.3.17 Valid datasets

Vamps recognizes the following datasets:[9]

pre precipitation [cm]

rlh relative humidity [

hea head at bottom [cm]

rdp rooting depth [cm]

tem dry bulb temp [oC]

gwt ground-water table [cm]

inr interception [cm]

trf troughfall [cm]

stf stemflow [cm]

pev potential evaporation [cm]

spe potential soil evaporation [cm]

ptr potential transpiration [cm]

qbo flow through bottom of profile [cm]

vol actual water content [cm]

[9] Used in the ts section and the - -Output option.

avt average theta in profile

lai leaf area index

sca canopy storage [cm???]

ira incoming radiation [W/m2]

nra net radiation [W/m2]

ref reflected radiation [W/m2]

win windspeed [m/s]

sur sunratio [n/N]

6.4 Interception

Vamps can determine interception loss via several methods. The Calder method (Calder 1986), the Gash analytical method (see paragraph 3.3.1) and Rutter's (Rutter et al. 1971) numerical method. For the Rutter method you will need precipitation data with hourly or shorter intervals. All parameters needed for interception calculation are set in the interception section of the input file. Each method has it's own set of required and optional arguments. In addition you also need at least one time-series (precipitation). Figure **??** and Figure **??** show a part of a input file which calculates interception using the Gash and the Calder method respectively. A description of all allowed variables in the interception section is given in section 6.3 on page 46.

Chapter 7

Running Vamps

Although you don't have to use the canopy module it is recommend that you do so if possible. The other methods of determining transpiration, interception etc. don't have a close interaction with the soil modules and usually provide poorer results.

7.1 Command line options

Options may be given as either standard one-letter options, or GNU style long options. Long options start with two dashes '' and may be truncated to any unambiguous abbreviation.

--Header Omit the headers in output.

--copyright Show copyright information and exit

--license Print license information and exit.

--help Print help text and exit.

--verbose Enter verbose mode, program progress is displayed on stderr. Default is silent.

--noinputdump By default the contents of the inputfile is also included in the output. Use this option to alter this behaviour.

--Output *setname* Dump the specified *setname* to a seperate file in **ts**(5) format. See **vamps**(5) for a list of allowed setnames.

--Determine *variable* Determine variable (can be a single variable or a time series). See **vamps**(5) for a list of allowed variables.

--Logfile *filename* Log this session to the file *filename*.

--Setvar *section name value* Set the value of variable *name* in section *section* to value *value*. This overrides the setting in the input file.

--Comment *commentchar* Set the commentchar to char *commentchar*.

--showdef Send program defaults to stdout. This can be used to create a default file from scratch.

--output *filename* Send output to *filename* in stead of stdout.

filename Read input from *filename*

7.2 Manipulation of time series

Links to the ts*(1) stuff

7.3 graphically displaying Vamps results

Links to the graph(1) stuff

7.4 Troubles..

OK, you have problems running Vamps. You are not alone! I will describe some problems that you can encounter using Vamps. Some have an easy solution, others not.

7.4.1 Variable missing

If Vamps issues a message like:

```
deffile.c: Fatal: could not find ->steps<-
 in section  ->time<-
 in file   ->example1.inp<-
deffile.c: error message:
 from: $Id: deffile.c,v 1.23 1996/03/18 09:24:50 schj Exp $
 description: Var not found or invalid, see above.
```

you have not specified a variable in the input file which Vamps *needs* to run properly. This can be a variable which is needed always or one that is needed is you have specified so in an other variable. For example, if you specify the Gash method for determining interception you *must* also specify the variable `S` in the `interception` section.

7.4.2 Mass-balance error warning

Vamps issues a warning when large jumps in the mass-balance error are encountered. You can ignore them providing you keep in mind that the solution is not very optimal. At present the mass-balance is *not* calculated properly is you use lateral drainage, beware! In most circumstances decreasing the `dtmax` variable helps, it prevents Vamps from making mistakes in estimating the max timestep. Descreasing the `dtmin` and/or the `thetol` variable should be next on your list followed by increasing the `maxitr` variable.

Chapter 8

Using S-Lang with Vamps

8.1 Introduction

To allow users to change functions that Vamps uses to determine like converting incoming radiation to net radiation as well as other emperical equations, Vamps includes the **S-Lang** stack based interpreter. Davis (1995b) describes **S-Lang** as:

> **S-Lang** (pronounced "sssslang") is a powerful stack based interpreter that supports a C-like syntax. It has been designed from the beginning to be easily embedded into a program to make it extensible. **S-Lang** also provides a way to quickly develop and debug the application embedding it in a safe and efficient manner. Since **S-Lang** resembles C, it is easy to recode **S-Lang** procedures in C if the need arises.

The actual syntax of **S-Lang** is not described here – see (Davis 1995b, Davis 1995a) for that – but the way Vamps uses **S-Lang** for it's extensions. Also Vamps specific **S-Lang** functions and variables are described here.

8.2 How Vamps interfaces with S-Lang

For speed reasons Vamps only uses the **S-Lang** interpreter at specific locations in the program:

1. at startup Vamps loads a file (usually `vamps.sl`) with the function definitions of functions that Vamps uses.
2. at the start of the program – after the input file is read – Vamps executes the `at_start` function
3. after each timestep the `each_step` is executed
4. after all calculations are done – but before data is flushed from memory – the `at_end` function is executed
5. a series of regression equations and relations between variables can be programmed in **S-Lang**. Vamps checks is a certain **S-Lang** function exits and

executes that in stead of the build in function. at present the following functions can be programmed in **S-Lang**:

- `lai_to_s (lai)` – returns s (canopy storage) for a given lai (leaf area index)
- .....

Chapter 9

Vamps specific functions and variables

9.1 Intrinsic functions (programmed in C)

9.2 User defined functions (written in S-Lang)

Part III

Case studies

Chapter 10

Introduction

Several studies in which the Vamps model has been used are presented in the following chapters. It is hoped that this not only gives a good idea about Vamps' capabilities, but will also help the user when working with the program. To give an idea of the possible uses of vamps I will shortly describe three (semi-realistic) cases and how Vamps can cope with these cases:

1. A small lysimeter located in west Java, Indonesia worked most of the time. However discharge data from several days is missing. Transpiration, interception en soil-evaporation fluxes are already calculated (and measured) by other means. In this case Vamps is used to estimate the discharge for the missing days. To simulate the lysimeter as accurate as possible Vamps is used with a no flow bottom boundary and lateral flow allowed only on the bottom soil node.

2. Fiji ...

Chapter 11

A *Pinus caribaea* plantation forest

11.1 Introduction

This chapter shows the results of running Vamps on a dataset concerning a *Pinus caribaea* plantation forest, located on Viti Levu, Fiji (see Waterloo (1994)). A 61 days period (July 2 – September 2) from the Tulasewa site was modeled using meteorological data with a 30 minute resolution. The soils at this site have a bulk density ranging from 0.97 at the top to 1.07 g/cm at 1.2 m below the surface. Saturated hydraulic conductivity was determined for three soil layers from 29 core samples (See Table 11.1). Waterloo (1994) determined van Genuchten parameters for the topsoil (0 – 30cm) and the subsoil (30 – 150cm) using 12 and 17 samples respectively. These values (see Table 11.1) were used in the soil section of the Vamps model without any modification.

Transpiration was modeled using the Penman Montheith combination equation while interception was modeled using an adapted version of the Gash model[1]. Soil evaporation was modeled by using the Penman Montheith combination equation as well. It was assumed that net radiation at the forest floor was 3.5 % of that at the top of the canopy.

To verify model results measured values of moisture content were used. All values were obtained using a capacitance probe.

11.2 Results

The water balance summary for the modeled period is shown int Table 11.2. The modeled interception and transpiration values are similar to those presented by Waterloo

[1]This version of Gash is described in the Vamps model documentation

Soil layer	Porosity	Alpha	K_{sat}[cm/day]	n	Depth [cm]
1	0.6	0.061	1800	1.098	0 – 30
2	0.61	0.042	380	1.094	30 – 75
3	0.61	0.042	0.03	1.094	70 – 150

Table 11.1: Soil physical parameters from to Tulasewa site used as input for Vamps

Parameter	Value
Initial water volume of profile	71.765
Saturated water volume of profile	98.397
Total precipitation	25.767
Total transpiration	16.753
Total interception	5.261
Total soilevaporation	1.522
Total rootextraction	10.693
Total drainage	-0.008
Total surface_runoff	0.000
Initial storage	71.765
Final storage	80.197
Change in storage	-8.432
Percent mass-balance err	-0.039

Table 11.2: Summary of modeled water balance results for the period July 2 – September 7

(1994). Canopy resistance (R_s) was calculated using the relations derived by Waterloo (1994)[2]. Calculated potential transpiration however, could not be maintained over this period due to modeled water stress. Calculated root-extraction (actual modeled transpiration) was 5.5 cm lower. This may be caused by several factors: *(1)* The regression equations used to estimate R_s were obtained for a wetter period in which the trees did not suffer water stress. *(2)* The water content versus suction-head functions are not very accurate in the drier regions, and may be overestimating the suction-head in those regions.

The average moisture content is generally modeled with good result (Figure 11.2, bottom part). The the upper layers the low moisture contents (theta ≤ 0.3). One reason for this is the the pF related to a moisture content of 0.3 is already beyond 4.2.

```
[vamps]
iniinmem=1

[run]
outputfile = run5.hh.out

[determine]
canopy = 1
soilmoisture = 1

[time]
steps = 2930

[ts]
precipitation=../ninp/precip.prn
netrad=../ninp/rnet.prn
rhumid = ../ninp/rh.prn
temp=../ninp/newt.prn
windspeed = ../ninp/wind.prn
sm_probe_avg_theta = ../ninp/theta.mes.ts

[xout]
filename= xout.arn

[interception]
method = 0
# These are pre-cyclone values
E_avg/R = 0.147
p_f = 0.6
p_tr = 0.017
S = 0.08
St = 0.0062

[canopy]
transpiration = 2
Rnet_absorb = 0.975
method = 2
layers = 1
#ra=7.0
z = 12.7
z_0 = 1.5
d = 7.0

[roots]
swsink = 0
swhypr = 0
swupfu = 0
depth = 120.0
hlim1 = -5.0
hlim2u = -50.0
hlim2l = -50.0
hlim3h = -800.0
hlim3l = -1000.0
hlim3 = -1800.0
hlim4 = -12000.0

[soil]
swredu=0
```

[2]See Example **??** which actually lists this function

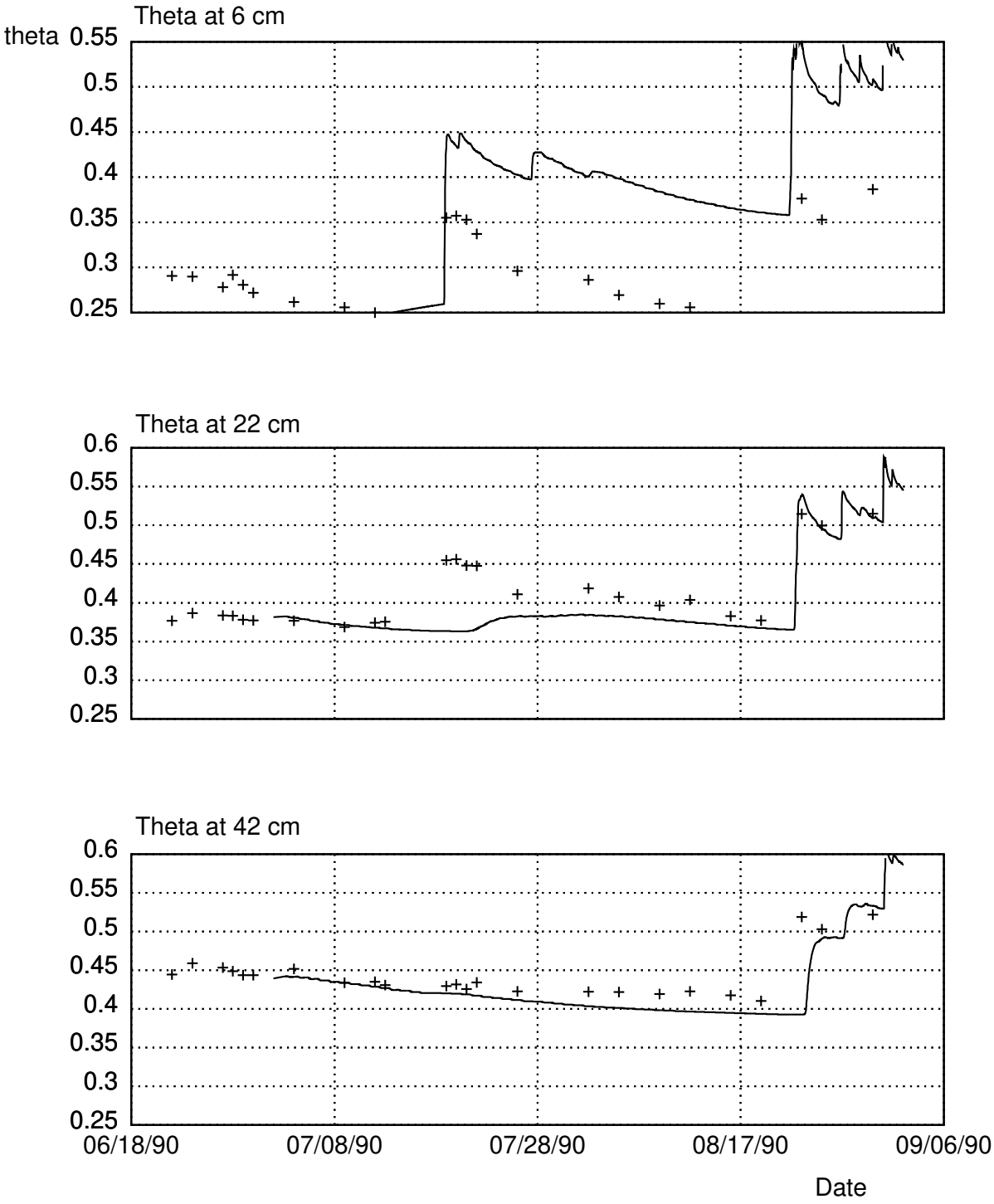


Figure 11.1: Measured (plus signs) and modeled (line) water content for layers at 6, 22 and 42 cm. below the surface

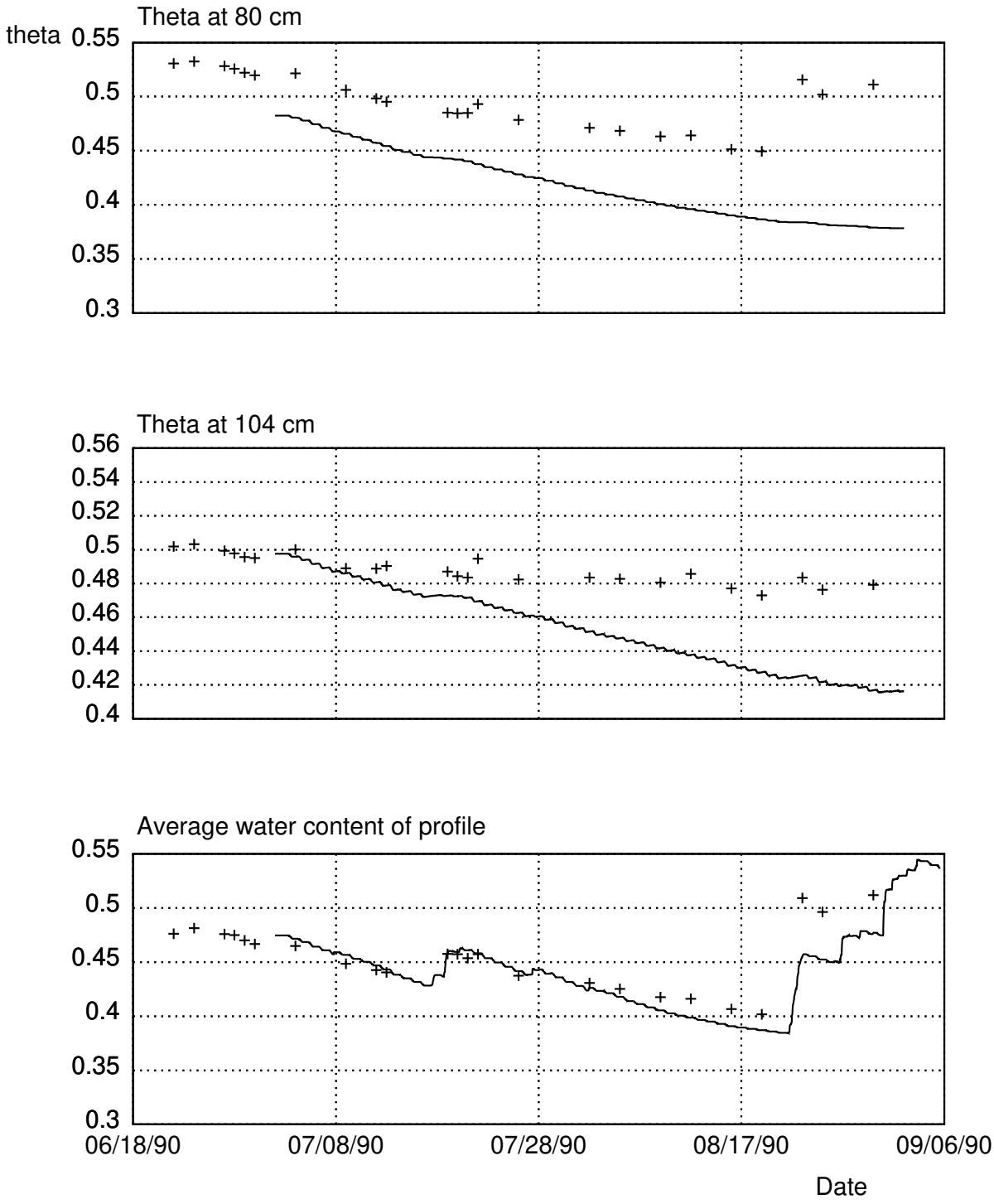


Figure 11.2: Measured (plus signs) and modeled (line) water content for layers at 80 and 104 cm. The bottom figure shows measured and modeled average water content.

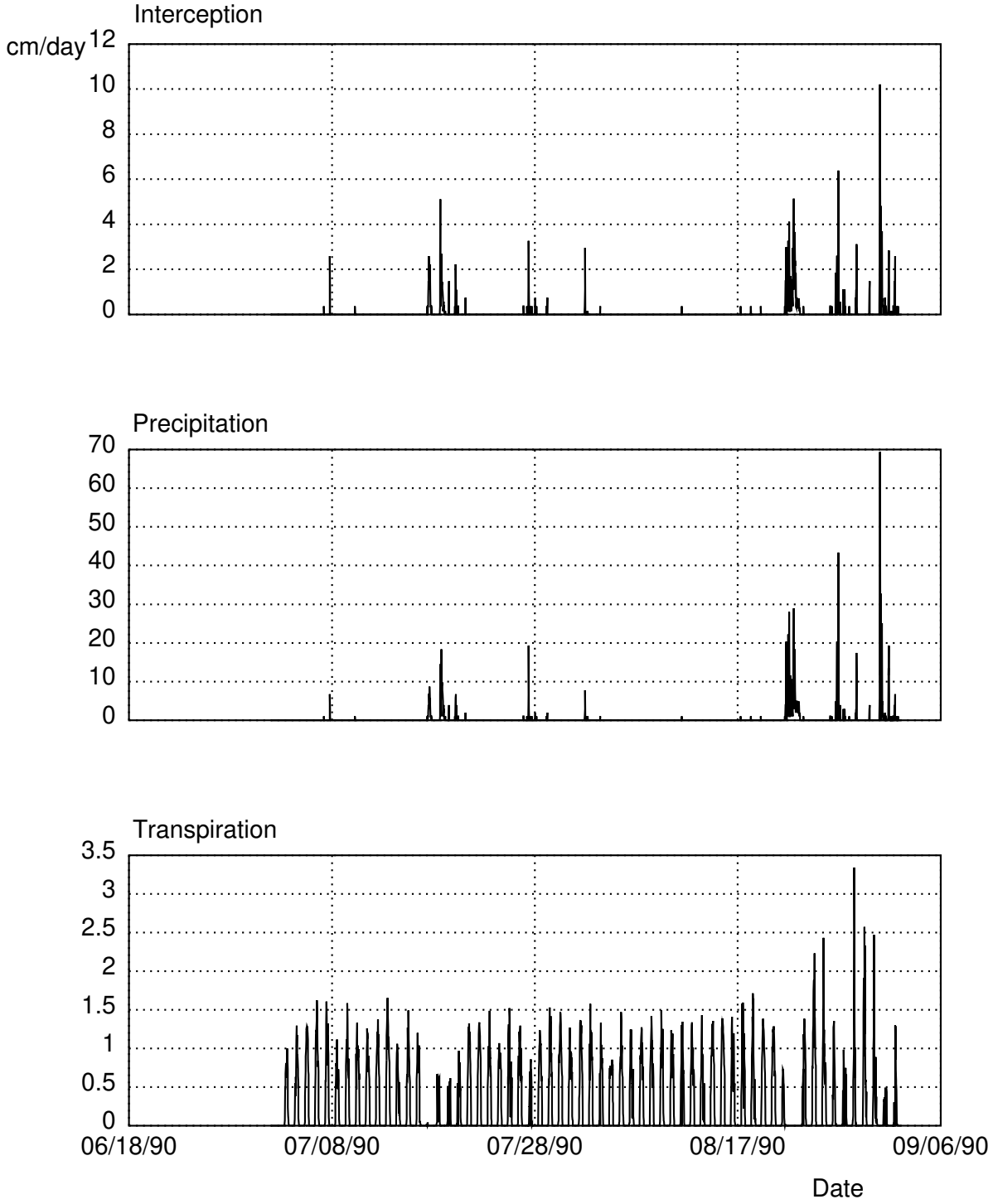

Interception
cm/day
12
10
8
6
4
2
0
Precipitation
70
60
50
40
30
20
10
0
Transpiration
3.5
3
2.5
2
1.5
1
0.5
0
06/18/90
07/08/90
07/28/90
08/17/90
09/06/90
Date

Figure 11.3:

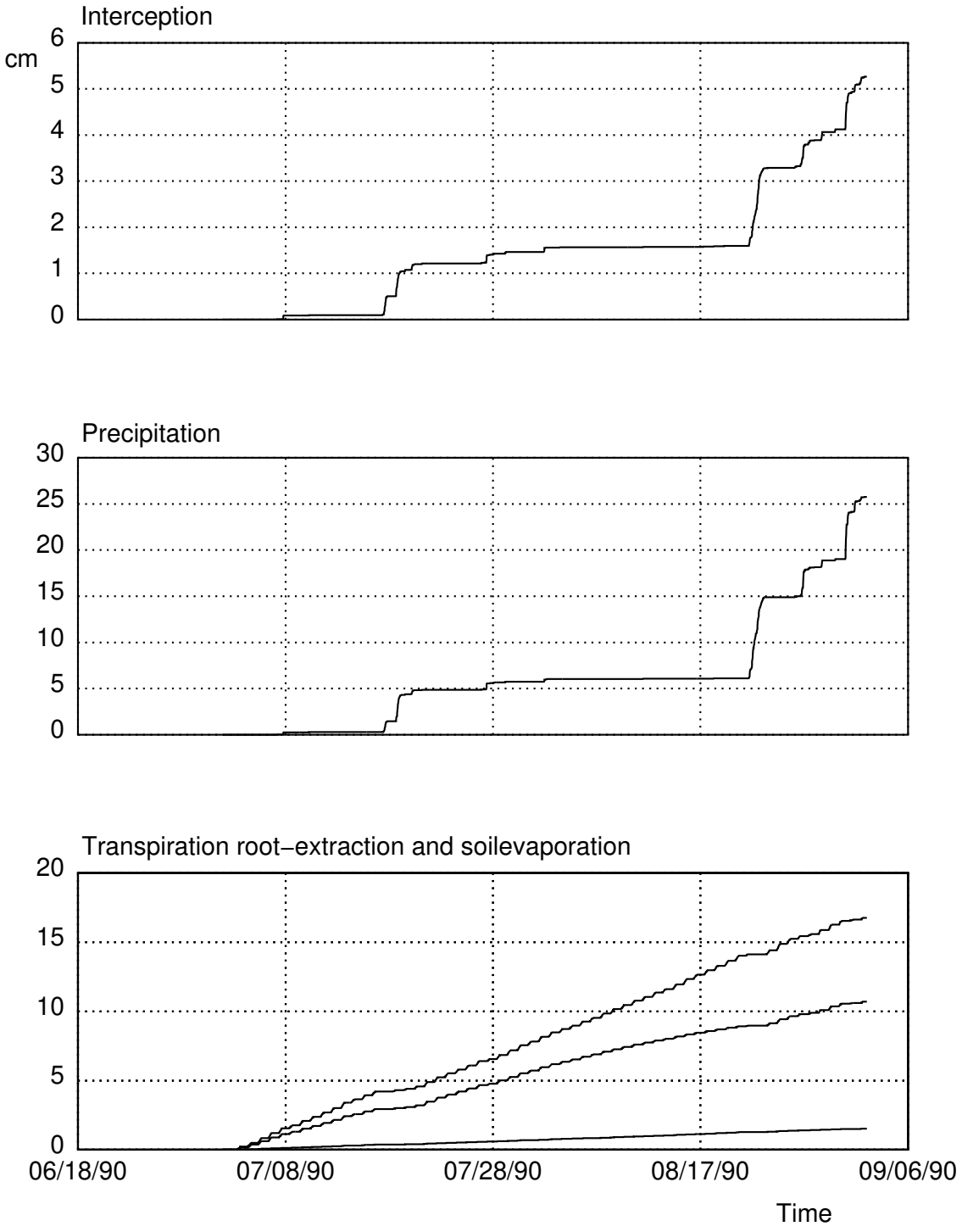

Interception
cm
6
5
4
3
2
1
0
Precipitation
30
25
20
15
10
5
0
Transpiration root−extraction and soilevaporation
20
15
10
5
0
06/18/90
07/08/90
07/28/90
08/17/90
09/06/90
Time

Figure 11.4:

```
cofred= 0.35
smooth=12
gwlevel = 1.0
outdir = output
pondmx = 0.0
verbose=1
layers= 77
bottom= 6
initprof = 0
theta_initial = 0.200000 0.210000 0.220000 0.220000 0.220000 0.220000
0.300000 0.350000 0.375817 0.377095 0.383497\
0.380000 0.380000 0.380000 0.443651 0.400000\
0.400000 0.400000 0.400000 0.410000 0.420000\
0.430000 0.440000 0.470000 0.470000 0.470000\
0.470000 0.470000 0.544194 0.470000 0.470000\
0.460000 0.460000 0.460000 0.460000 0.582024\
0.460000 0.460000 0.460000 0.460000 0.460000\
0.460000 0.514537 0.506907 0.506545 0.505096\
0.504733 0.495707 0.489597 0.492829 0.501117\
0.505820 0.499311 0.497508 0.497508 0.497508\
0.497508 0.497508 0.497508 0.497508 0.497508\
0.497508 0.497508 0.497508 0.497508 0.497508\
0.497508 0.497508 0.497508 0.497508 0.497508\
0.497508 0.497508 0.497508 0.497508 0.497508\
0.497508 0.497508

# This is from page 70 phd waterloo
# total depth = 2.000000
[layer_0]
description = Tulasewa top layer
thickness=2.000000
theta_initial=0.2
method =1
thetas =0.6
theta_residual = 0.08
thickness = 2.5
alpha = 0.061
n = 1.098
l=0.5
ksat=0.0

# total depth = 30.000000
[layer_14]
description = Tulasewa 30 - 75 cm layer
thickness=2.000000
theta_initial=0.4
thetas = 0.64
alpha = 0.042
n = 1.094
l = 0.5
ksat=380.0

# total depth = 74.000000
[layer_36]
description = Tulsewa deep layer > 75 cm
thickness=2.000000
theta_initial=0.46
ksat = 3.0
thetas = 0.6
```

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Index

C, 25, 29
D_r, 29
E_0, 23
E_a, 24
E_c, 29
E_i, 27
LAI, 28
P, 27, 28
P', 27
R_s, 24
R_{net}, 24
R_{nl}, 24
Rs_{out}, 24
S, 27
T, 25
Δ, 24
α, 24
δ, 28
γ, 24, 28
λ, 25
$\overline{E}$, 27
$\overline{R}$, 27
e, 25
e_a, 24
e_s, 24
$laifrac$, 28
n/N, 25
p, 27
p_{Tr}, 27
u, 24
–prefix=/your/installdir, 37
/usr/local, 37
0install, 37
0readme, 37

alpha, 52
arrays, 44

boolean, 44
bottom, 34

Calder regression model, 28
capacitance probe, 67
characters, 44
commentchar, 44
configure, 37

dpmi, 40
drainage, 51
drootz, 47
dtmax, 58
dtmin, 30, 58

estdmc, 51

firststep, 41
floats, 44

Gash, 27
getstr, 39
gnuplot, 39
graph, 39

h2dmc, 51
hb = -16.4, 53
HOME, 37

integers, 44
Interception, 26
interception, 27, 58
Interception as a fraction of LAI, 28
issec, 39

layer_0, 53
layer_n, 41, 52
layers, 41

maxitr, 30, 58
mktable, 51, 52

n, 52

operating systems, 40

pf, 52

Rutter's model, 28

S, 58
setenv LDFLAGS -L/home/schj/lib, 37
slope, 51
soil, 34, 41, 50, 52
soillib.inp, 33
soilsection, 52
src, 37
steps, 45
strings, 44

tablesize, 50
theta, 52
thetol, 30, 58
time, 41
ts, 27, 47

vamps, 44
vamps.rc, 37
vamps.sl, 37
vsel, 39

Word Wide Web, 17
wrini, 39